

Generative Imaging using Diffusion Models

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References

The coding examples are adopted from:

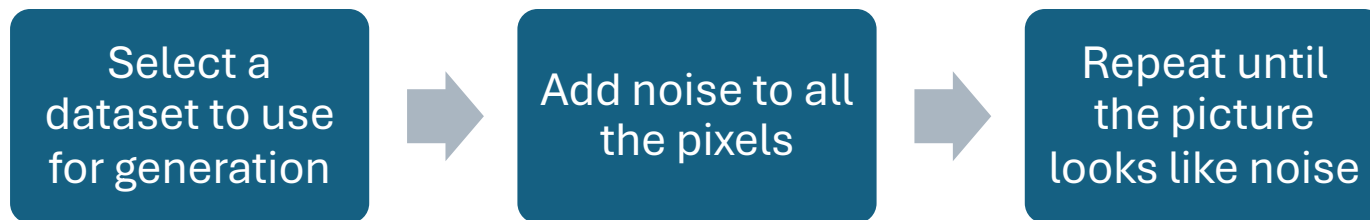
A. Géron (2025). *Hands-On Machine Learning with Scikit-Learn and PyTorch: Concepts, Tools, and Techniques to Build Intelligent Systems*. O'Reilly.

The basic **learning models** are from:

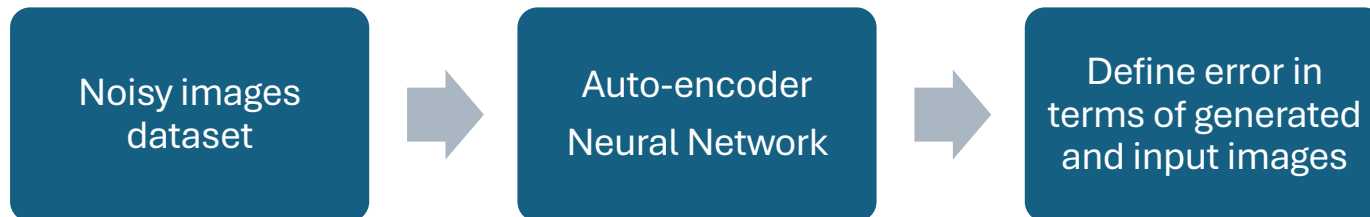
Torralba, A., Isola, P., & Freeman, W. T. (2024). *Foundations of Computer Vision*. The MIT Press. <https://visionbook.mit.edu/>.

The Diffusion Process: Training

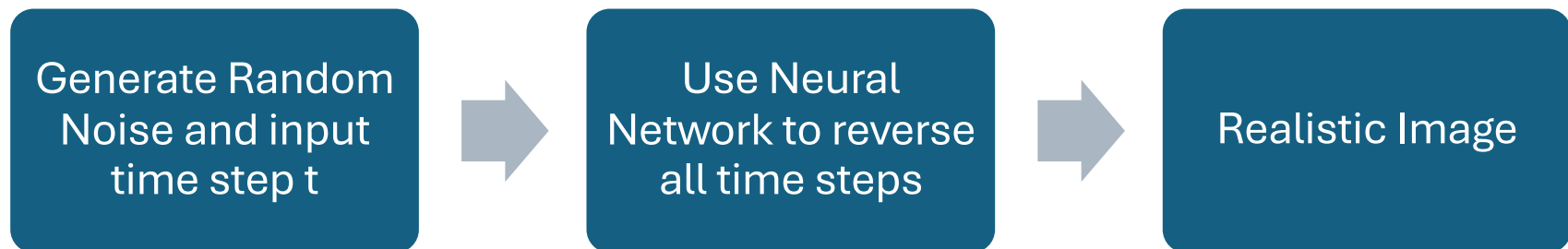
Step 1. Build a dataset to train the diffusion process:



Step 2. Train an auto-encoder neural network to remove noise from an image:



The Diffusion Process: Deployment



Training dataset: Create a sequence from any input image

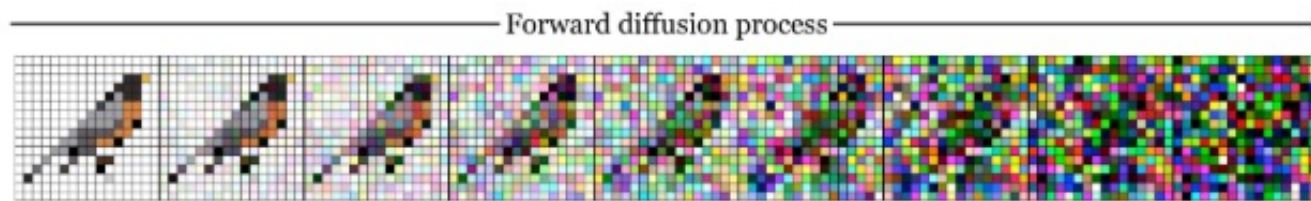


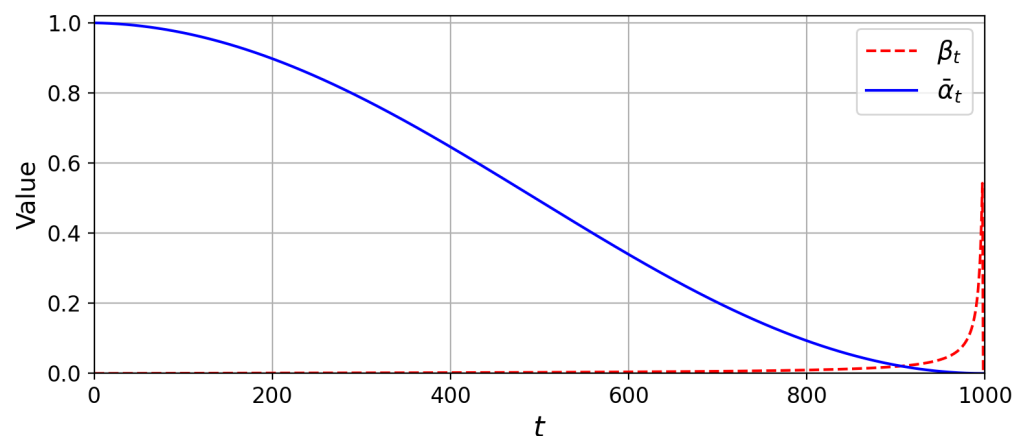
Figure 32.14: Forward diffusion process.

After T steps of this process, the image $\mathbf{x}_T$ looks like pure noise, if T is large enough. In fact, if we use the following noise process, then $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ as $T \rightarrow \infty$ (which follows from equation 4 in [4]).

$$\begin{aligned}\mathbf{x}_t &= \sqrt{(1 - \beta_t)}\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon_t \\ \epsilon_t &\sim \mathcal{N}(\mathbf{0}, \mathbf{I})\end{aligned}$$

Generate noisy videos

Code uses:
 $ab = 1 - \beta_t$
 $\beta_t = 1 - ab$



$$\mathbf{x}_t = \sqrt{(1 - \beta_t)}\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon_t$$

$$\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$$

Signal coefficient keeps going down.
 Noise coefficient picks up at the end.

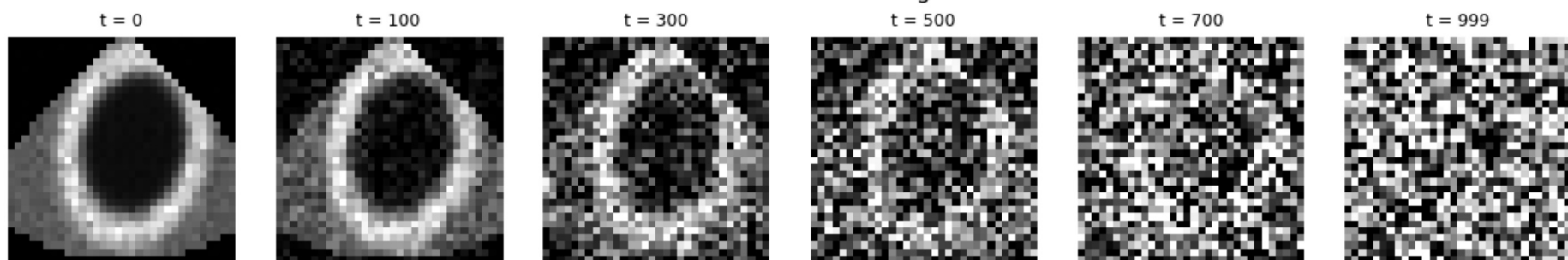
```
N_STEPS = 1000                                # number of noise levels: t = 0, 1, ..., 999
s = torch.arange(N_STEPS + 1, device=device) / N_STEPS
alpha_bar = (torch.cos((s + 0.008) / 1.008 * math.pi / 2) ** 2) # cosine schedule
alpha_bar = (alpha_bar[1:] / alpha_bar[0]).clamp(1e-5, 1)        # 1 = clean ... ~0 = pure noise

def add_noise(x0, t, noise):
    ab = alpha_bar[t].view(-1, 1, 1, 1, 1) # one alpha_bar per video, shaped to match x0
    return ab.sqrt() * x0 + (1 - ab).sqrt() * noise # x_t = sqrt(ab)*x0 + sqrt(1-ab)*noise

# In the training loop:
noise = torch.randn_like(x0) # random Gaussian noise, same shape as the videos
xt = add_noise(x0, t, noise)  # noisy videos at random noise levels t
```

Adding noise until it looks like noise

First frame of one video at increasing noise levels



Neural Net Goal: Reverse the sequence

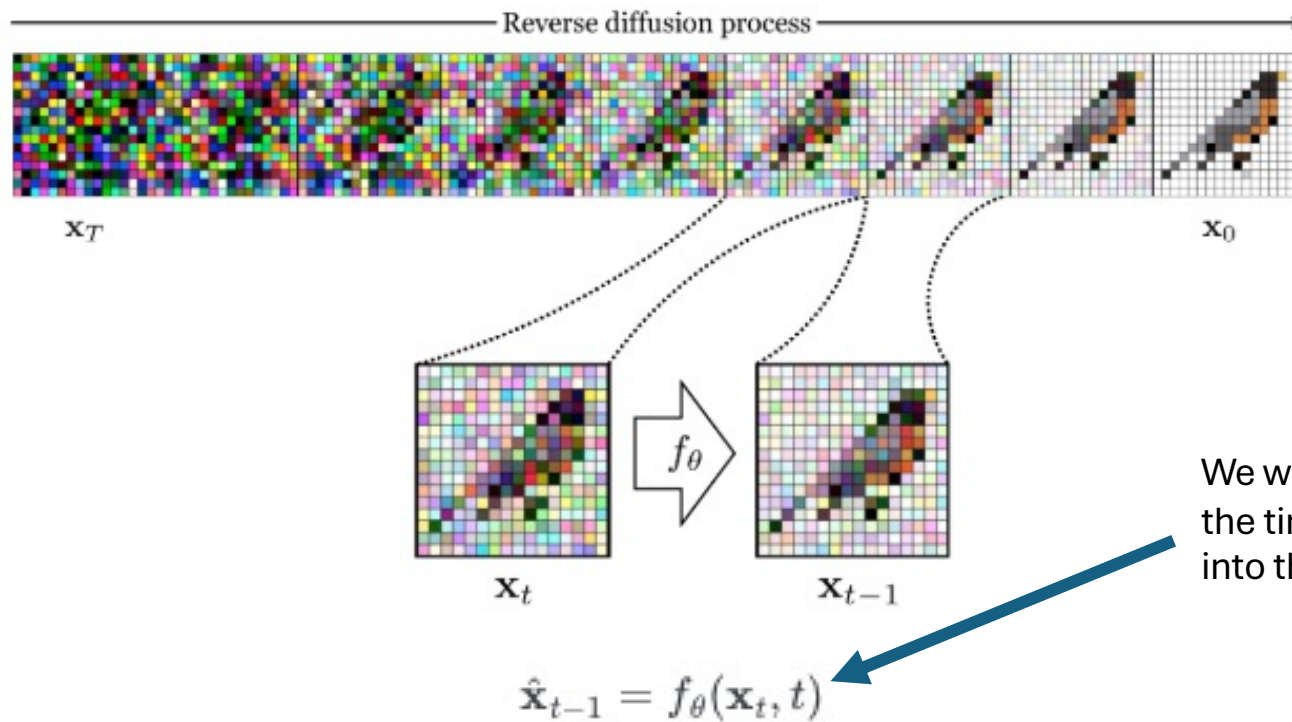


Figure 32.15: Reverse diffusion process.
The forward process creates supervision to train the reverse process.

We will look at embedding the time t and other parameters into the input.

<https://visionbook.mit.edu/>

Use an Auto-encoder as a Diffusion Model

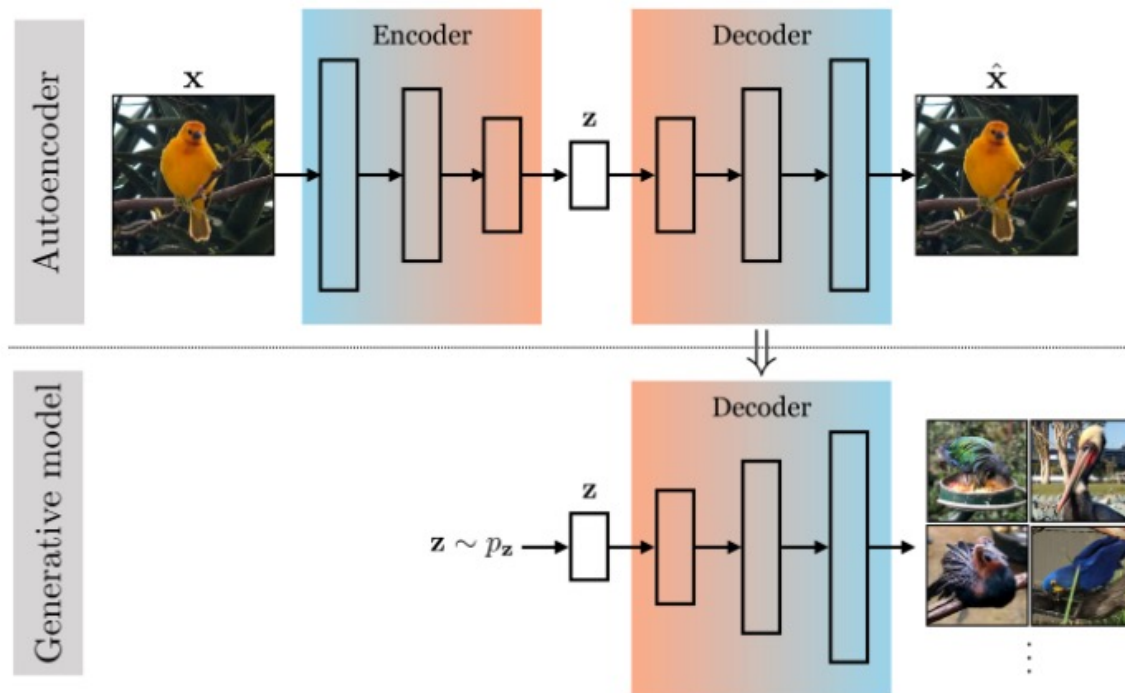


Figure 33.3: The relationship between an autoencoder and a generative model.

Embedding Time using sinusoids

```
def timestep_embedding(t, dim):  
    half = dim // 2  
    freqs = torch.exp(-math.log(10000) * torch.arange(half, device=t.device) / half)  
    args = t.float()[ :, None] * freqs[None]  
    return torch.cat([args.sin(), args.cos()], dim=1)
```

Residual block: from c_in to c_out channels

The skip connection converts from input channels to output channels if needed, enabling a residual connection.

```
class ResBlock(nn.Module):
    def __init__(self, c_in, c_out, emb_dim):
        super().__init__()
        self.norm1, self.conv1 = nn.GroupNorm(8, c_in), nn.Conv3d(c_in, c_out, 3, padding=1)
        self.emb = nn.Linear(emb_dim, c_out)
        self.norm2, self.conv2 = nn.GroupNorm(8, c_out), nn.Conv3d(c_out, c_out, 3, padding=1)
        self.skip = nn.Conv3d(c_in, c_out, 1) if c_in != c_out else nn.Identity()
    def forward(self, x, e):
        h = self.conv1(F.silu(self.norm1(x)))
        h = h + self.emb(e)[: , :, None, None, None]
        h = self.conv2(F.silu(self.norm2(h)))
        return h + self.skip(x)
```

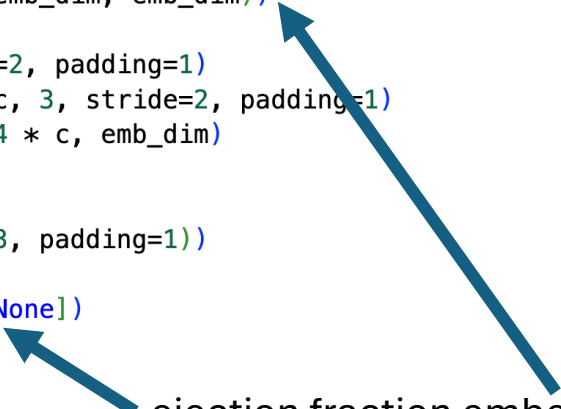
Residual connection.

ejection fraction embedding propagation through linear layer.

UNet3D: Autoencoder with time and ef embeddings

```
class UNet3D(nn.Module):
    def __init__(self, c=16, emb_dim=128):
        super().__init__()
        self.emb_dim = emb_dim
        self.t_mlp = nn.Sequential(nn.Linear(emb_dim, emb_dim), nn.SiLU(), nn.Linear(emb_dim, emb_dim))
        self.ef_mlp = nn.Sequential(nn.Linear(1, emb_dim), nn.SiLU(), nn.Linear(emb_dim, emb_dim))
        self.inc = nn.Conv3d(1, c, 3, padding=1)
        self.d1, self.down1 = ResBlock(c, c, emb_dim), nn.Conv3d(c, c, 3, stride=2, padding=1)
        self.d2, self.down2 = ResBlock(c, 2 * c, emb_dim), nn.Conv3d(2 * c, 2 * c, 3, stride=2, padding=1)
        self.mid1, self.mid2 = ResBlock(2 * c, 4 * c, emb_dim), ResBlock(4 * c, 4 * c, emb_dim)
        self.u2 = ResBlock(4 * c + 2 * c, 2 * c, emb_dim)
        self.u1 = ResBlock(2 * c + c, c, emb_dim)
        self.out = nn.Sequential(nn.GroupNorm(8, c), nn.SiLU(), nn.Conv3d(c, 1, 3, padding=1))

    def forward(self, x, t, ef):
        e = self.t_mlp(timestep_embedding(t, self.emb_dim)) + self.ef_mlp(ef[:, None])
        up = lambda h: F.interpolate(h, scale_factor=2, mode="nearest")
        h1 = self.d1(self.inc(x), e) # 16 x 32 x 32
        h2 = self.d2(self.down1(h1), e) # 8 x 16 x 16
        h = self.mid2(self.mid1(self.down2(h2), e), e) # 4 x 8 x 8
        h = self.u2(torch.cat([up(h), h2], 1), e)
        h = self.u1(torch.cat([up(h), h1], 1), e)
        return self.out(h) # predicted v, same shape as x
```



ejection fraction embedding.

Training EMA using UNet3D

```
STEPS, BATCH, LR, EMA_DECAY = 1500, 32, 5e-4, 0.995
```

```
use_amp = device == "cuda"
```

```
opt = torch.optim.AdamW(model.parameters(), lr=LR)
```

```
ema = copy.deepcopy(model).eval().requires_grad_(False)
```

```
scaler = torch.amp.GradScaler(enabled=use_amp)
```

ema model.

```
losses, t0 = [], time.time()
```

```
model.train()
```

```
for step in range(1, STEPS + 1):
```

Add noise and train UNet3D to reverse.

```
    x0, ef = make_echo(BATCH)
```

```
    t = torch.randint(0, N_STEPS, (BATCH,), device=device)
```

```
    noise = torch.randn_like(x0)
```

```
    ab = alpha_bar[t].view(-1, 1, 1, 1)
```

```
    v_target = ab.sqrt() * noise - (1 - ab).sqrt() * x0
```

```
    with torch.autocast(device_type=device, dtype=torch.float16, enabled=use_amp):
```

```
        loss = F.mse_loss(model(add_noise(x0, t, noise), t, ef).float(), v_target)
```

```
    opt.zero_grad(set_to_none=True)
```

```
    scaler.scale(loss).backward()
```

```
    scaler.step(opt); scaler.update()
```

```
    with torch.no_grad():
```

```
        for p_ema, p in zip(ema.parameters(), model.parameters()):
```

```
            # EMA update
```

```
            p_ema.lerp_(p, 1 - EMA_DECAY)
```

Train ema model.

EMA is an **exponential moving average of the model's weights**. In the diffusion notebook, a second copy of the U-Net slowly follows the one being trained, and that smoothed copy is the one used to **generate** the videos.

The code

```
ema = copy.deepcopy(model).eval().requires_grad_(False)    # a frozen copy of the U-Net
...
# after every training step:
for p_ema, p in zip(ema.parameters(), model.parameters()):
    p_ema.lerp_(p, 1 - EMA_DECAY)                          # EMA_DECAY = 0.995
```

`lerp_` means "linear interpolation, in place". For every weight:

$$w_{\text{EMA}} \leftarrow 0.995 \cdot w_{\text{EMA}} + 0.005 \cdot w_{\text{model}}$$

After each step, the EMA copy moves **0.5% of the way** toward the current weights. Older values fade gradually (exponentially), so the EMA copy is effectively an average of the weights over roughly the **last $1 / (1 - 0.995) = 200$ steps**.

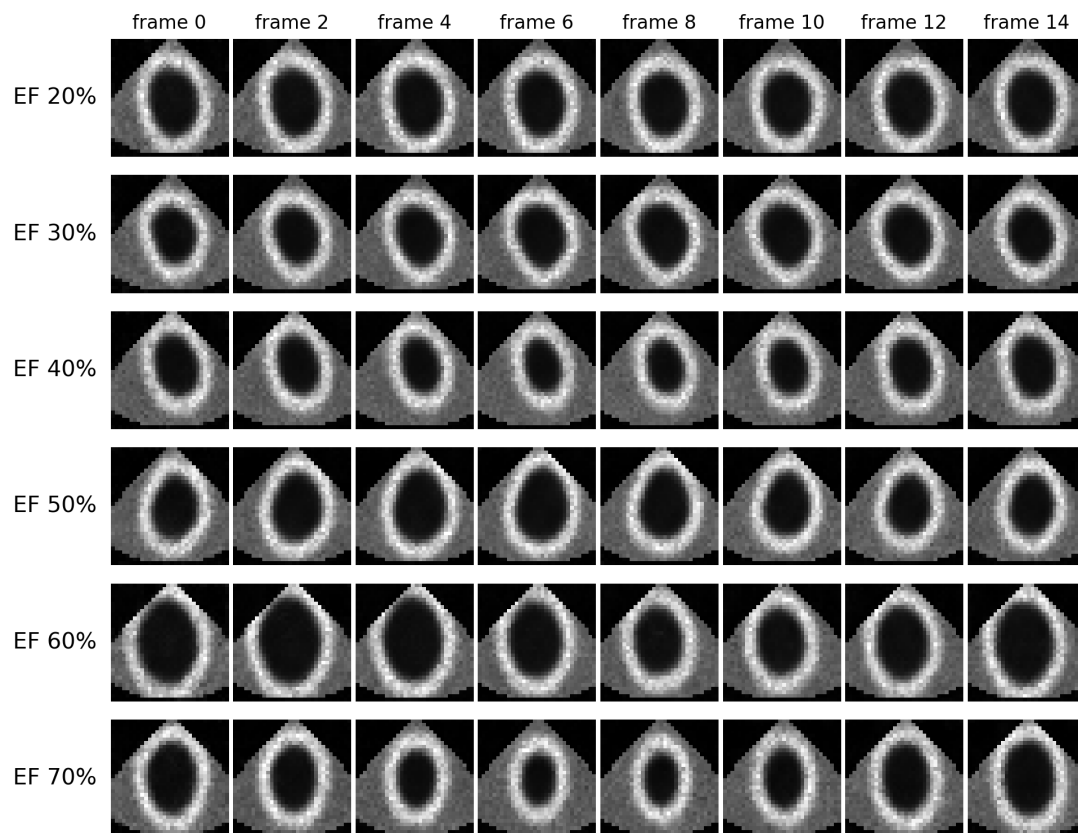
The EMA model is never trained directly: it gets no gradients (`requires_grad_(False)`), and the optimizer doesn't touch it. It only follows.

Use ema to generate the target with given ef

```
@torch.no_grad()
def generate(ef, steps=50, net=None):
    net = net or ema                                # use the EMA weights
    ef = ef.to(device)
    x = torch.randn(len(ef), 1, T, S, S, device=device) # start from pure noise
    ts = torch.linspace(N_STEPS - 1, 0, steps, device=device).long()
    for i, t in enumerate(ts):
        ab = alpha_bar[t]
        ab_next = alpha_bar[ts[i + 1]] if i + 1 < steps else torch.tensor(1.0, device=device)
        with torch.autocast(device_type=device, dtype=torch.float16, enabled=use_amp):
            v = net(x, t.repeat(len(x)), ef).float()
            x0 = (ab.sqrt() * x - (1 - ab).sqrt() * v).clamp(-1, 1) # estimate of the clean video
            eps = (x - ab.sqrt() * x0) / (1 - ab).sqrt()
            x = ab_next.sqrt() * x0 + (1 - ab_next).sqrt() * eps # DDIM step to the next level
    return x

t0 = time.time()
requested = torch.tensor([0.2, 0.3, 0.4, 0.5, 0.6, 0.7]) ← Generate these ejection fractions.
fake = generate(requested)
print(f"Generated {len(fake)} videos in {time.time() - t0:.1f} s")
animate(fake, [f"generated, EF {e:.0%}" for e in requested])
```

Example Output: Heart Videos

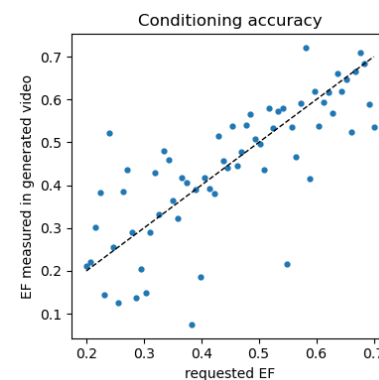


Ejection fraction (EF): the fraction of blood the left ventricle pumps out with each heartbeat.

EF = (EDV - ESV) / EDV EDV, ESV: ventricle volume when most filled / most emptied

Normal: about 55–70%. 40% or less: reduced (heart failure).

In our videos (2D): EF = 1 - smallest cavity area / largest cavity area



Above: requested EF vs. EF measured in 64 generated videos.

Correlation 0.77, mean error 0.07

Do the videos appear realistic? Does each heart pump as requested?